WASABI (Input output / one timepoint version)

Simplified formulation

$$\mathbf{r} \sim \text{Poisson}(\boldsymbol{\lambda})$$
$$\boldsymbol{\lambda}^{\text{stim}} = \mathbf{C}\mathbf{s} + \boldsymbol{\lambda}^{\text{control}}$$
$$\mathbf{C} = \mathbf{D}\mathbf{W}_{\text{low-rank}}$$

$\mathbf{r}$:	Recorded responses (spikes)
$\boldsymbol{\lambda}$:	Firing rates
$\mathbf{C}$:	Mapping from stimulation channels to firing rates
$\mathbf{s}$:	Selection of stimulation channels
$\mathbf{D}$:	Mapping from latent variables to recorded responses
$\mathbf{W}_{\text{low-rank}}$:	Low rank mapping from stimulation channels to latent variables

- $\mathbf{W}$ is unknown.
- At time t-1, we want to find $\mathbf{s}_t$ that maximizes the information about $\mathbf{W}$, given $\mathbf{r}_{1:t-1}$ and $\mathbf{s}_{1:t-1}$

Ridge regression for learning W at time t

If we assume we know D_t at time t, we estimate $\mathbf{W}_t$ using a ridge least-squares objective:

$$\hat{\mathbf{W}}_t = \arg\min_{\mathbf{W}} ||\mathbf{Z} - \mathbf{WS}||_F^2 + \beta||\mathbf{W}||_F^2$$

where β is the weight of the ridge prior. We can expanding the Frobenius norm using trace:

$$\mathcal{L}(\hat{\mathbf{W}}_t) = tr[(\mathbf{Z} - \mathbf{WS})(\mathbf{Z} - \mathbf{WS})^\top] + \beta\, tr(\mathbf{WW}^\top)$$
$$\mathcal{L}(\hat{\mathbf{W}}_t) = tr(\mathbf{ZZ}^\top) - 2\, tr(\mathbf{ZS}^\top\mathbf{W}^\top) + tr(\mathbf{WSS}^\top\mathbf{W}^\top) + \beta\, tr(\mathbf{WW}^\top)$$

Keep terms relative to $\mathbf{W}_t$

$$\mathcal{L}(\hat{\mathbf{W}}_t) = -2\,\mathrm{tr}(\mathbf{ZS}^\top\mathbf{W}^\top) + \mathrm{tr}(\mathbf{WSS}^\top\mathbf{W}^\top) + \beta\,\mathrm{tr}(\mathbf{WW}^\top)$$

Taking derivative and set to zero:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \mathbf{W}} &= -2\,\mathbf{ZS}^\top + 2\,\mathbf{WSS}^\top + 2\beta\,\mathbf{W} \\
0 &= -2\,\mathbf{ZS}^\top + 2\,\mathbf{WSS}^\top + 2\beta\,\mathbf{W} \\
\mathbf{WSS}^\top + \beta\mathbf{W} &= \mathbf{ZS}^\top \\
\mathbf{W}(\mathbf{SS}^\top + \beta\mathbf{I}) &= \mathbf{ZS}^\top
\end{aligned}$$

$$\boxed{\mathbf{W} = \mathbf{ZS}^\top(\mathbf{SS}^\top + \beta\mathbf{I})^{-1}}$$

Generative model for perturbation

$$(\boldsymbol{\lambda}_{\text{stim}} - \boldsymbol{\lambda}_{\text{control}}) = \mathbf{D}(\mathbf{WS} + \boldsymbol{\xi}) + \boldsymbol{\eta}$$

Note:

- Current loss is written for estimating $\mathbf{W}_t$
- $\mathbf{D}$ - learned from control? or learn jointly with trials t (Alternateing or SVD)?
- $\boldsymbol{\xi}_t$ is perturbation noise (e.g. mismatch between intended and actual stim-response effects) - hyperparameter, match data?
- $\boldsymbol{\eta}$: observation noise - hyperparameter, match data?
- Columns of $\mathbf{W}$ have unit variance
- Ridge prior should scale as $\beta \propto \frac{1}{n}$ so in low-trial regime solution is not degenerate and high trial regime data info is respected.

Minimizing the posterior covariance of W (Gaussian)

For adaptive selection of stimulation patterns, at each trial t, we want to select $\mathbf{s}_t$ that maximizes the expected information gain about $\mathbf{W}$. Equivalently, this would be selecting $\mathbf{s}_t$ that minimizes the determinant of the covariance of $\mathbf{W}$ at time t. We separate out the Poisson MLE rate estimation and identification of the optimal $\mathbf{W}^*$ in our formulation. This formulation could be good for control-theoretic implementations later too.

$$\boldsymbol{\lambda}_t = \mathbf{D}\mathbf{W}\mathbf{s}_t + \boldsymbol{\lambda}_t^{control}$$

Here, we are trying to find the optimal $\mathbf{W}^*$, where $\boldsymbol{\lambda}$, $\mathbf{s}_t$, and $\boldsymbol{\lambda}_t^{control}$ are known.

Let Σ_{t-1}	be the covariance matrix of $\mathbf{W}$ before trial $t-1$
Let β	be how we scale our data relative to our prior (could be $1/n$)

In terms of precision, we can formulate the Gaussian case as:

$$\Sigma_t^{-1} = \Sigma_{t-1}^{-1} + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^T$$

Take the determinant

$$\det \Sigma_t^{-1} = \det(\Sigma_{t-1}^{-1} + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^T)$$

using the matrix inversion lemma:

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}^T) = (1 + \mathbf{v}^T\mathbf{A}^{-1}\mathbf{u})\det(\mathbf{A})$$

$$\det(\Sigma_{t-1}^{-1} + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^T) = (1 + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^T\Sigma_{t-1}\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t])\det(\Sigma_{t-1}^{-1})$$

Since we minimize over $\mathbf{s}_t$, the objective function is therefore:

$$\begin{aligned}\arg\max_{\mathbf{s}} \det\left(\Sigma_t^{-1}\right) &= \arg\max_{\mathbf{s}} \left(1 + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^\top\Sigma_{t-1}\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\right) \det(\Sigma_{t-1}^{-1}) \\ &= \arg\max_{\mathbf{s}} \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^\top\Sigma_{t-1}\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\end{aligned}$$

We can further expand out:

$$\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t] = \hat{\mathbf{D}}_{t-1}\hat{\mathbf{W}}_{t-1}\mathbf{s}_t + \boldsymbol{\lambda}_t^{control}$$

where $\hat{\mathbf{D}}_{t-1}$ and $\hat{\mathbf{W}}_{t-1}$ are estimated from timepoints $i = 1, \ldots, t-1$ and $\boldsymbol{\lambda}_t^{control}$ is firing rate identified from control reaches.

Need to find the correct way to express how s affects the uncertainty of W Then take the nuclear norm and then max min (E optimality) max min singular value of precision